

Why Remote Work Stuck

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Motivation: The Remote Work Puzzle

- Remote work **doubled** after COVID-19
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 - Zoom daily meeting: 10M in 2019 → 300M in 2020 (↑ 2,900%).
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- Despite being a valued amenity, remote jobs are associated with **higher wages**

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We build a structural model to decompose their contributions and explain wage patterns.

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- Estimate a **search and matching model** with:
 - Workers and firms jointly choose remote intensity
 - Heterogeneity in skills, preferences, and productivity
- **Quantitatively decompose** preference vs. technology channels

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- **Preference shifts** account for roughly **57%** of the expansion.
- **Technology improvements** explain about **33%**.
- The remaining **10%** is explained by **changes in skill returns and the skill distribution** across workers.

Literature

I. Teleworkability Measurement

- Dingel and Neiman (2020), Mongey, Pilossoph and Weinberg (2021)
- **Contribution:** We build a **continuous measure of occupational "teleworkability"** expand binary classification to a continuous index.

II. Remote Work and Compensation

- Barrero, Bloom, Buckman, and Davis (2023), Pabilonia and Vernon (2023), Cullen, Pakzad-Hurson and Perez-Truglia (2024)
- **Contribution:** Link wage premiums to occupation teleworkability.

III. Amenity Provision and Hedonic Wages

- Hwang, Mortensen and Reed (1998), Morchio and Moser (2024)
- **Contribution:** Link worker preferences for remote work to job productivity. Allow for heterogeneity in remote work efficiency at the firm and worker level.

Roadmap

1. Data & Teleworkability Measure
2. Three Motivating Facts
3. A Model of Remote Work Choice
4. Results & Decomposition
5. Conclusion

Data Sources

- **Current Population Survey (CPS)** - Labor Market Outcomes
 - 2019 - 2024 Workers aged 25-64
 - *Key variables:* Wages, remote work share, occupation, demographics

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- **Teleworkability Index Construction:**
 - **BLS Occupational Requirements Survey (ORS)**: Ground truth for telework feasibility
 - **O*NET Database**: 200+ occupation characteristics (tasks, skills, context)
 - *Method:* Train ML model (ORS + O*NET) → Predict teleworkability index for all occupations

Measuring Teleworkability

- **Prior work:** Binary classification ([Dingel & Neiman, 2020](#))
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 - Remote efficiency varies across occupations

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 - Jobs are either teleworkable or not: $\{0, 1\}$
- **Why we need a continuous measure?**
 - Hybrid work exists ($\sim 10\%$ of jobs)
 - Remote efficiency varies across occupations
- **Example:**

Occupation	D&N	Our Index
Security Guards	0	0.00
Historians	0	0.34

Measuring Teleworkability

- **Data:**
 - **Occupation Characteristics** O*NET data on tasks and skills.
 - **Occupation Requirements Survey** Telework percentages from a subset of occupations.
- **Two-stage approach:**
 - Stage 1 (Extensive): Classifier → predict if job can be done remotely.
 - Stage 2 (Intensive): Regressor → assign continuous teleworkability score.

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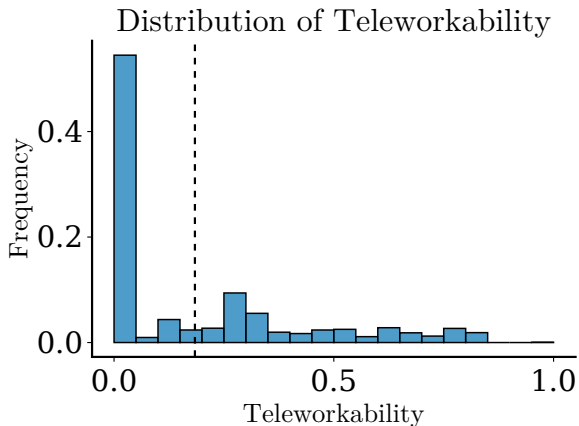
- Stage 1 (Extensive): Classifier → predict if job can be done remotely.
- Stage 2 (Intensive): Regressor → assign continuous teleworkability score.

- **Performance:**

- Stage 1: $F1 = 0.915$: strong model performance, with both false positives and false negatives.
- Stage 2: Correlation = 0.832: accurate continuous prediction.

Measuring Teleworkability: Results

Occupation	D&N	Ours
Security Guards	0	0.00
Clinical Lab Techs	0	0.00
Industrial Managers	0	0.12
Historians	0	0.37
Tax Preparers	0	0.30
Economists	1	0.59
Lawyers	1	0.83
Web Developers	1	1.00



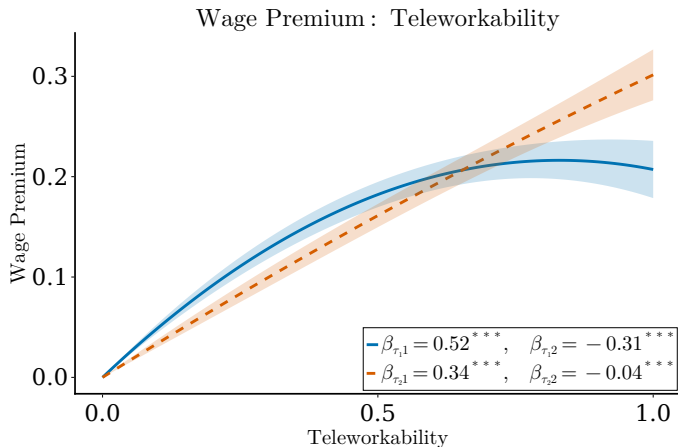
- Total 879 occupations classified.

Fact 1: Higher teleworkability higher wages

[Full Table](#)

$$\log(w_{i,t}) = \sum_{D \in \{\tau_1, \tau_2\}} \mathbb{1}[t \in D]_i \times \left(\beta_{D1} \psi_{i,t} + \beta_{D2} \psi_{it}^2 + \gamma_{D1} \alpha_{i,t} + \gamma_{D2} \alpha_{it}^2 \right) + X'_{i,t} \Gamma + \phi_t + \varepsilon_{i,t}$$

- τ_1 : Pre-Pandemic
- τ_2 : Post-Pandemic
- ψ : Teleworkability Index
- α : Remote Work

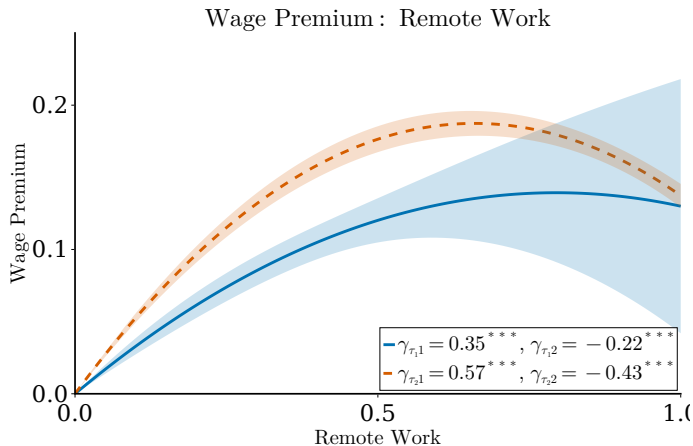


Fact 2: Remote Work Wage Premium

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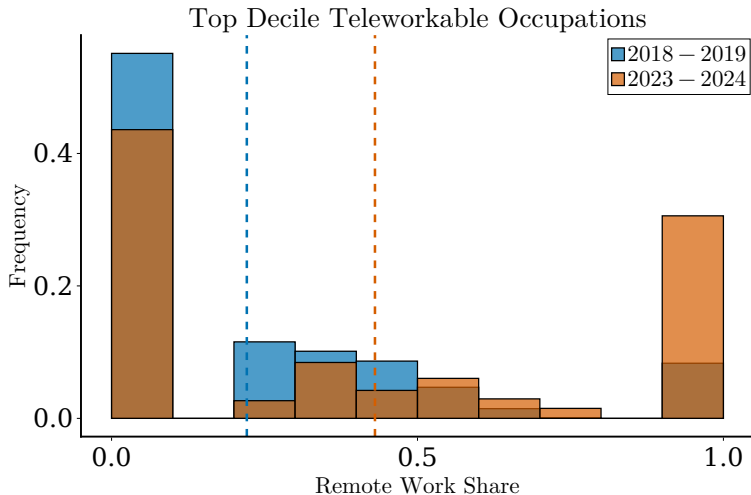


Fact 3: Hybrid Work is Common in Teleworkable Jobs

- Mass in the interior is $\approx 30\%$

- Dispersion Within Jobs

- Top Quartile



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 - ψ is permanent and observed by all agents.
- **The Choice:** Worker and firm jointly choose:
 - Remote work share $\alpha \in [0, 1]$ to maximize joint surplus.
 - Wage w to split the surplus.
- **Search Process:** Random search with standard matching frictions.
 - No on-the-job search—workers.

Production Technology

- Output depends on skill (h), firm efficiency (ψ), and remote share (α).

$$Y(\alpha \mid \psi, h) = Ah \cdot [(1 - \alpha)^\rho + \alpha^\rho \cdot g(\psi)]^{1/\rho}$$

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- **Relative Productivity**: The term $g(\psi) = \psi_0 \cdot \psi^\nu$ is the remote technology component.
 - ψ_0 : Baseline productivity of remote work.
 - ν : How remote productivity scales with firm efficiency ψ .

Worker Preferences

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- χ : Curvature of amenity value (e.g., diminishing returns if $\chi > 0$)
- The amenity is scaled by a match-specific **taste parameter** z .
 - z is drawn from a known distribution F_Z when the worker and firm meet.

The Core Trade-Off: Building the Joint Surplus

- For any match, the firm and worker choose the remote share α to maximize their combined total surplus.

$$\alpha^* \in \operatorname{argmax}_{\alpha \in [0,1]} \left\{ \underbrace{\text{Production}(\alpha, h, \psi) - w}_{\text{Firm's Profit}} + \underbrace{w + \text{Amenity Value}(\alpha)}_{\text{Worker's Utility}} \right\}$$

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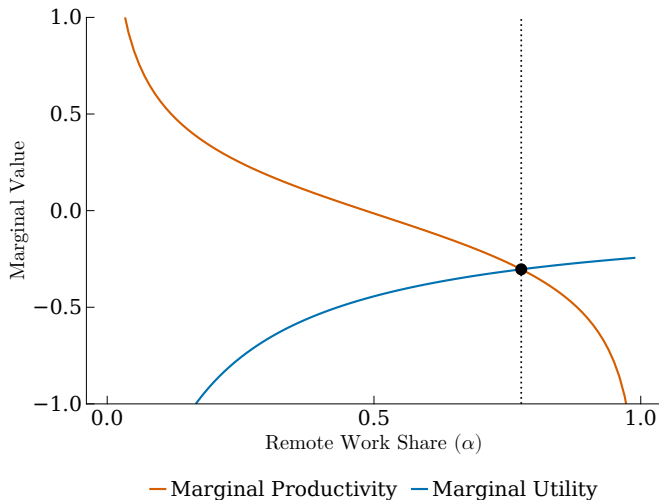
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The first-order condition reveals the core trade-off in the model:

$$\underbrace{\frac{\partial Y}{\partial \alpha}}_{\text{Marginal Product of Remote Time}} + \underbrace{\frac{\partial A}{\partial \alpha}}_{\text{Marginal Amenity Value of Remote Time}} = 0$$

Optimal Work Arrangement: The Trade-off

$$-\frac{\partial Y}{\partial \alpha} = \frac{\partial A}{\partial \alpha}$$

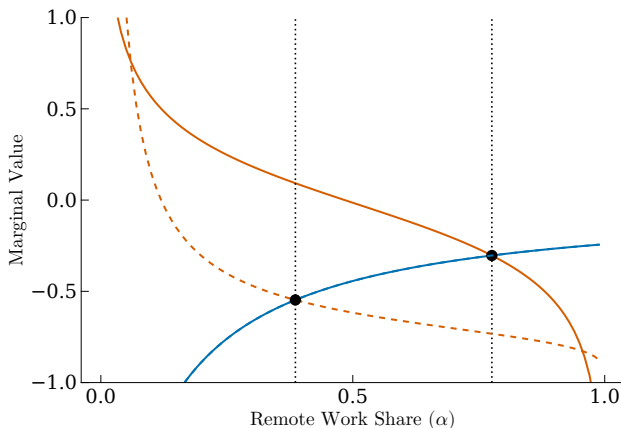


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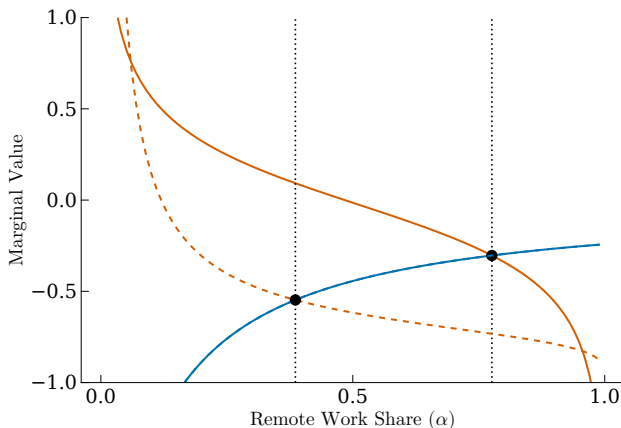
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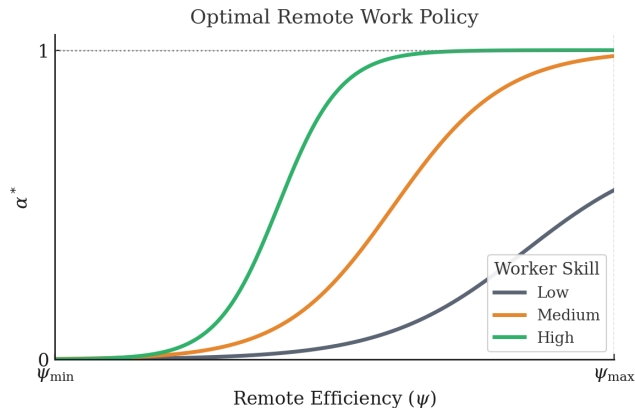
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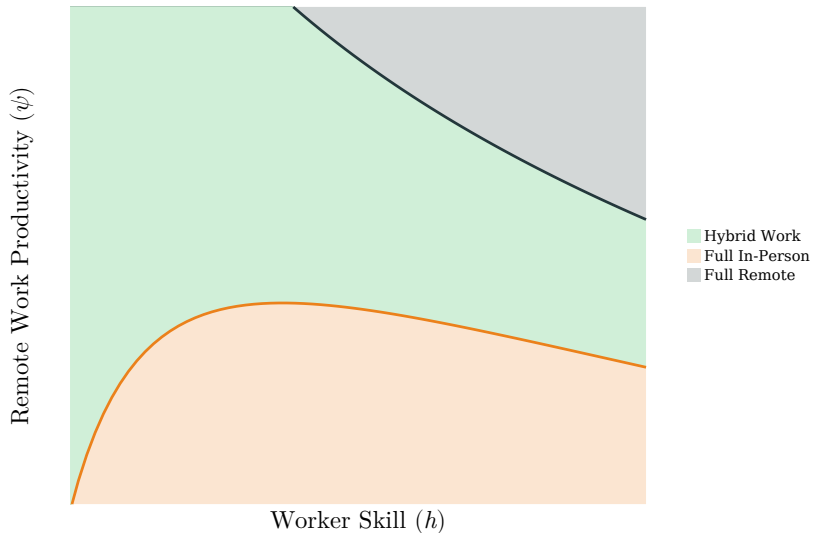
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Optimal $\alpha^*(\psi)$

- Increasing in h and ψ



Work Arrangement



Match Surplus: The Role of α

1. **Given** z : Choose α^* to maximize joint value

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2. Flow surplus is the **expected** optimal value

$$s(h, \psi) = \mathbb{E}_z \left[\max_{\alpha} \{ Y(\alpha \mid \psi, h) + z \cdot \mathcal{A}(\alpha) \} \right] - b(h)$$

Match Surplus: Bellman Equation

Total match surplus:

$$S(h, \psi) = \underbrace{s(h, \psi)}_{\text{Flow Surplus}} + \underbrace{\beta(1 - \delta)S(h, \psi)}_{\text{Continuation}} - \underbrace{\beta p(\theta) \xi \int \max\{0, S(h, \psi')\} d\Gamma_v(\psi')}_{\text{Search Cost}}$$

- $p(\theta)$: job-finding rate
- $\Gamma_v(\psi')$: equilibrium vacancy distribution

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- **Wage:** Nash bargaining determines how surplus is split

$$w^*(h, \psi; \alpha^*) = \underbrace{(1 - \beta(1 - \delta))[U(h) + \xi S(h, \psi)] - \beta\delta U(h)}_{\text{Base Wage}} - \underbrace{z \cdot \mathcal{A}(\alpha^*)}_{\text{Amenity Deduction}}$$

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- Smooth production–amenity trade-off.
- This generates a sizable interior region of hybrid matches in equilibrium.

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- We estimate these distribution parameters along with technology and search friction parameters to match key moments in the data.

Identification: Mapping Moments to Parameters

Parameter(s)	Target Moment	Intuition
<i>Worker Skill Distribution</i>		
a_h, b_h	Mean of log wages	Central tendency
	Variance of log wages	Wage dispersion
<i>Preferences (Taste Shocks)</i>		
μ_z	Wage gap (remote vs. in-person)	Location of taste distribution
χ	Share fully remote ($\alpha = 1$)	Preference curvature
σ_z	Share hybrid ($0 < \alpha < 1$)	Taste dispersion
<i>Production Technology</i>		
ν	Wage- ψ slope	Firm-remote complementarity
ψ_0	Wage premium (high- ψ firms)	Productivity scale
	$\mathbb{E}[\alpha \mid \psi_{\text{high}}] - \mathbb{E}[\alpha \mid \psi_{\text{low}}]$	
<i>Search Frictions</i>		
κ_0	Market tightness (V/U ratio)	Vacancy cost

Parameter Estimates: A Structural Shift from 2019 to 2024

Parameter	2019	2024	Change (%)	Contrib.
Preferences				57%
χ (Amenity Curvature)	0.634	0.409	-35.49%	
μ_z (Preference Dist. Mean)	0.0088	0.0098	+12.29%	
σ_z (Preference Dist. Std Dev)	1.254	1.629	+29.89%	
Technology				33%
ψ_0 (Baseline Remote)	0.392	0.571	+45.66%	
ν (Firm-Technology)	0.032	0.039	+21.88%	
ρ (Input Substitutability)	0.999	0.999	0.00%	
Skill Distribution				10%
$(h \sim \text{Beta}(a_h, b_h))$				
a_h	4.912	4.209	-14.31%	
b_h	4.118	4.964	+20.54%	
TFP				—
A_1	14.197	16.448	+15.86%	

• Preferences Increased

- Preferences became more *flat*.
- Mean taste for flexibility went up.
- Taste dispersion increased.

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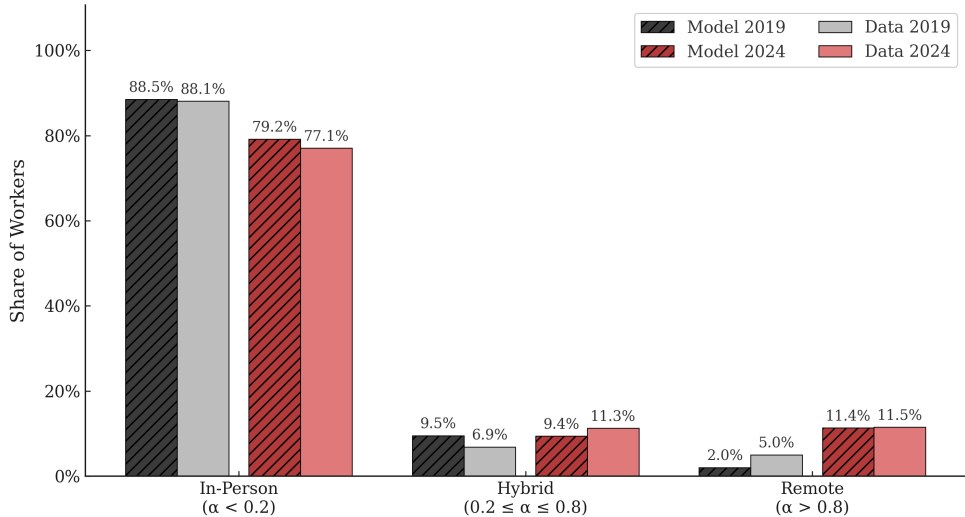
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- **Preference Increased**
- **Technology Improved:**
 - $\uparrow \psi_0$ everyone became more efficient at remote work.
 - $\uparrow \nu$ the improvement is larger for high ψ .
 - Not a noticeable change in ρ .

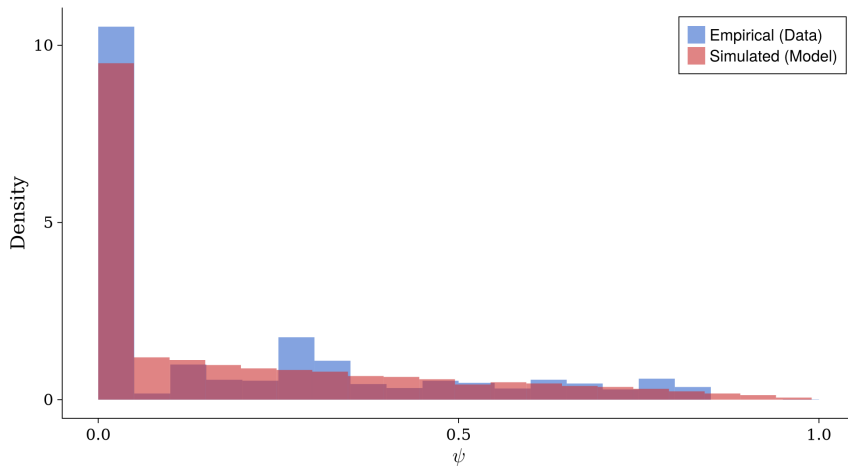
Model Fit

Moment	2019		2024	
	Data	Model	Data	Model
Mean Log Wage	3.028	3.046	3.105	3.104
Variance of Log Wage	0.416	0.493	0.341	0.462
P90/P10 Log Wage Ratio	1.607	1.716	1.530	1.627
Mean ψ	0.177	0.155	0.198	0.192
Mean α	0.064	0.068	0.178	0.135
Var α	0.019	0.012	0.094	0.079
Full In-Person Share	0.881	0.885	0.771	0.792
$\alpha - \psi$ Slope	0.186	0.275	0.490	0.802
Job Filling Rate	0.807	0.592	0.704	0.544
Wage- α Slope	0.108	-0.017	0.163	-0.141
Wage- α Curvature	-0.022	-1.182	-0.598	-0.314
Wage- ψ Slope	0.615	1.45	0.656	1.70
Wage- ψ Curvature	-0.282	-1.182	-0.04	-0.314

Decomposition of Work Arrangement Changes



Distribution of Teleworkability (ψ)



Empirical: $\mu=0.193$, $\sigma=0.247$ | Simulated: $\mu=0.195$, $\sigma=0.248$

Decomposition of Changes in Remote Work

1. Total Change to Explain:

- $16.0\% - 6.6\% = 9.4\%$.
- Represents 100%.

Decomposition $\bar{\alpha}$: 2019→2024



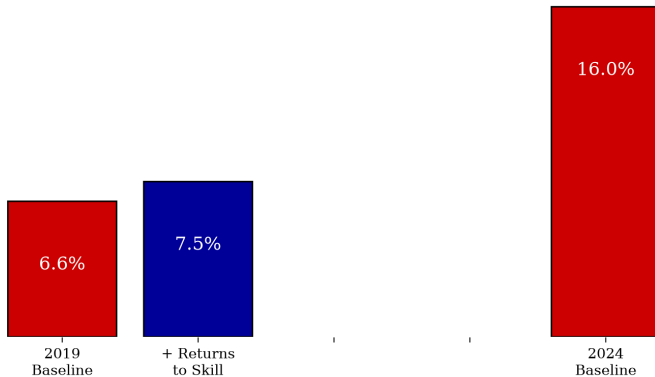
Decomposition of Changes in Remote Work

1. Total Change to Explain:

- $16.0\% - 6.6\% = 9.4\%$.
- Represents 100%.

2. + Returns to Skill 9.6%

Decomposition $\bar{\alpha}$: 2019→2024



Decomposition of Changes in Remote Work

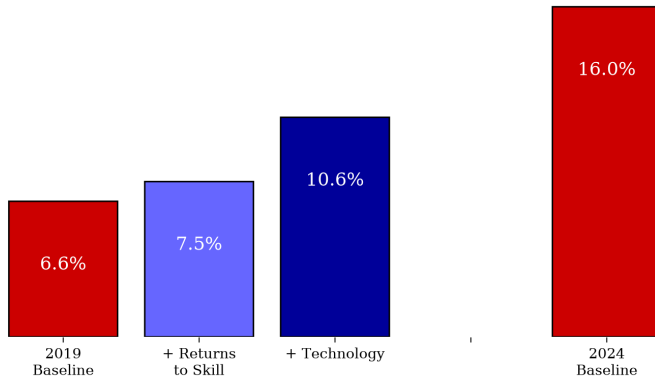
1. Total Change to Explain:

- $16.0\% - 6.6\% = 9.4\%$.
- Represents 100%.

2. + Returns to Skill 9.6%

3. + Technology 33.0%

Decomposition $\bar{\alpha}$: 2019→2024



Decomposition of Changes in Remote Work

1. Total Change to Explain:

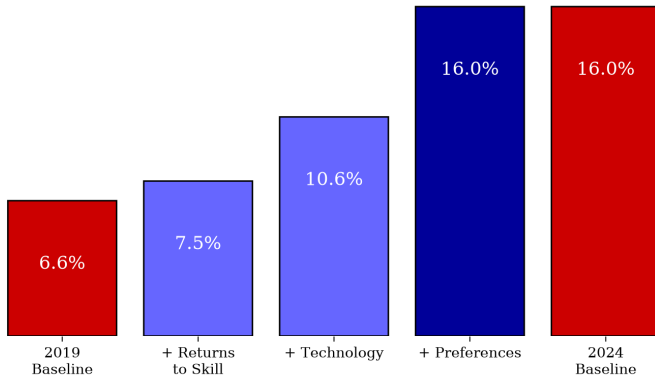
- 16.0% - 6.6% = 9.4%.
- Represents 100%.

2. + Returns to Skill 9.6%

3. + Technology 33.0%

4. + Preferences 57.4%

Decomposition $\bar{\alpha}$: 2019→2024



Conclusion

- **Research Question:** What drove the persistent rise in remote work post-pandemic?
- **Our Approach:**
 - Built a continuous measure of occupational teleworkability (ψ)
 - Developed a structural search model with heterogeneous workers and firms
 - Estimated the model separately for 2019 and 2024 to identify structural changes
- **Main Finding:** The rise in remote work reflects **both** preference and technology channels:
 - **Preference shifts** account for **57%** of the increase
 - **Technology improvements** explain **33%**
- **Policy Implication:**
 - The shift to remote work is driven by genuine preference changes, not just technological adaptations.
 - Suggests it's likely to persist.
 -

Thank You!

O*NET Occupational Characteristics (Predictors):

- **Work Activities:** 41 task-level characteristics
 - Examples: Analyzing Data, Working with Computers, Face-to-Face Discussions
- **Work Context:** 57 environmental conditions
 - Examples: Physical Proximity, Deal with External Customers, Freedom to Make Decisions
- **Skills:** 35 worker capabilities
- **Abilities:** 52 worker capacities

BLS Occupational Requirements Survey (Training Labels):

- Survey of ~ 800 occupations (2023 wave)
- Direct measurement: “What percentage of workers in this occupation can telework?”
- Provides ground truth for $\psi \in [0, 1]$

Stage 1: Binary Classification

- **F1 Score:** 0.915

Stage 2: Continuous Prediction

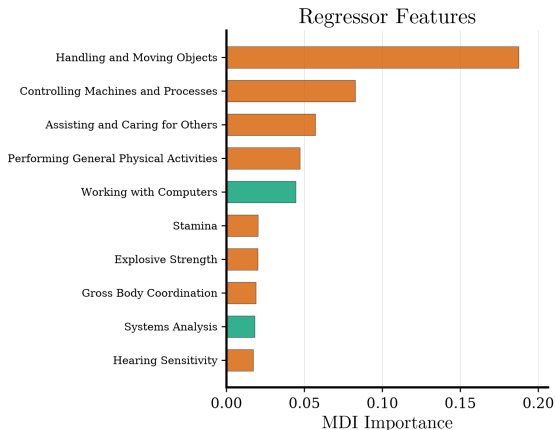
- **MAE:** 0.055 (5.5% error)
- **MAE (non-zero):** 0.116
- **Correlation:** 0.832

Model Selection: Random Forest chosen after testing 49 classifier-regressor combinations

Feature Importance

[Back](#)

- **Mean Decrease in Impurity (MDI):** higher $\implies$ classification more certain.



Green: increases teleworkability

Orange: decreases teleworkability

Appendix A4: Index Distribution & Examples

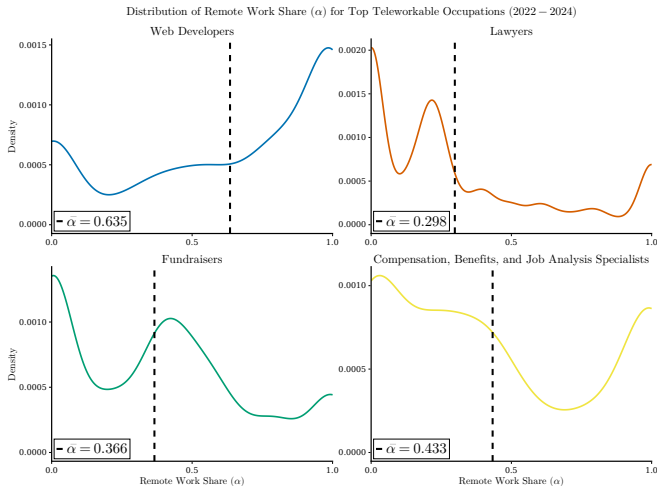
[Back](#)

Top 10 Occupations	Avg ψ	Sample of Occupations with $\psi = 0$
Web Programers	1.0	Construction Trades Workers
Lawyers, Judges, and Related Workers	0.824	Entertainment Attendants
Advertising, Marketing, PR, Sales Mgrs	0.695	Farming/Fishing Supervisors
Sales Reps, Wholesale & Manufacturing	0.605	Forest, Conservation Workers
Mathematical Science Occupations	0.590	Metal/Plastic Workers
Operations Specialties Managers	0.580	Other Teachers/Instructors
Architects, Surveyors, Cartographers	0.571	Protective Service Supervisors
Physical Scientists	0.569	Food Preparation Workers
Financial Specialists	0.519	Material Moving Workers
Sales Representatives, Services	0.478	Firefighting Workers
$\mathbb{E}[\psi \mid \psi > 0] = 0.215$		Share $\psi = 0$: 26.1%

Large Within-Occupation Dispersion

[Back](#)

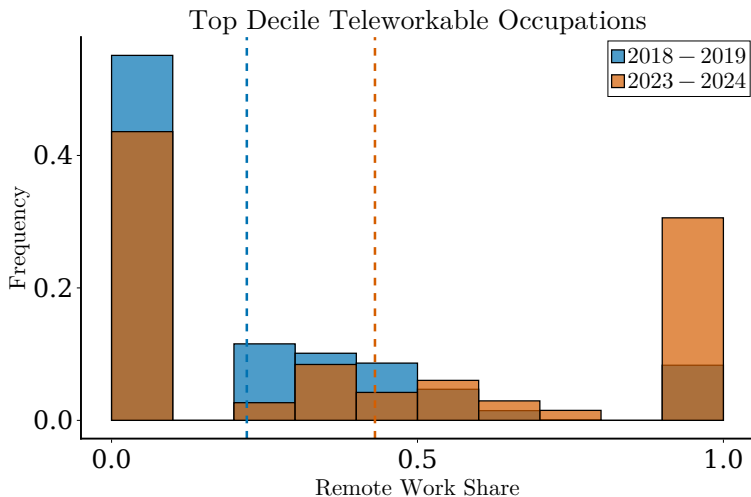
- Within a single, narrowly-defined occupation, we see a wide variety of work arrangements.
- **The Implication:** This is strong evidence for the role of **idiosyncratic taste heterogeneity** (z) in driving choices.



Hybrid Work is Common in Teleworkable Jobs

[Back](#)

- Mass in the interior is $\approx 30\%$



Flow Surplus Definition:

$$s(h, \psi) = \underbrace{\mathbb{E}_z \left[\max_{\alpha \in [0,1]} \{ Y(\alpha \mid \psi, h) + z \cdot A(\alpha) \} \right]}_{\text{Expected Match Value}} - \underbrace{b(h)}_{\text{Home Production}}$$

Home Production:

We assume the value of home production is proportional to the average total flow value a worker of that skill type could generate in the market:

$$b(h) = b \cdot \mathbb{E}_\psi[\pi(h, \psi)]$$

Bellman Equation for a Match ($J(h, \psi)$): The joint value of a match, $J(h, \psi)$, is the present discounted value of all future returns:

$$J(h, \psi) = s(h, \psi) + \beta [(1 - \delta)J(h, \psi) + \delta U(h)]$$

Bellman Equation for Unemployment ($U(h)$) The value of unemployment consists of the current flow benefit, $b(h)$, plus the expected value from job search:

$$U(h)(1 - \beta) = b(h) + \beta p(\theta) \xi \int \max\{0, S(h, \psi')\} d\Gamma_v(\psi')$$

Appendix: Wage Equation Derivation

[Back](#)

Step 1: Worker's Bellman Equation

The worker's value in a match includes wage, amenity value, and continuation:

$$W(h, \psi) = (w + z \cdot \mathcal{A}(\alpha^*)) + \beta(1 - \delta)W(h, \psi) + \beta\delta U(h)$$

Step 2: Solve for Wage

Rearranging to isolate w :

$$w = (1 - \beta(1 - \delta))W(h, \psi) - \beta\delta U(h) - z \cdot \mathcal{A}(\alpha^*)$$

Step 3: Apply Nash Bargaining

Nash bargaining gives: $W(h, \psi) = U(h) + \xi S(h, \psi)$. Substituting:

$$w^*(h, \psi; \alpha^*) = \underbrace{(1 - \beta(1 - \delta))[U(h) + \xi S(h, \psi)] - \beta\delta U(h)}_{\text{Base Wage (constant in } \alpha)} - \underbrace{z \cdot \mathcal{A}(\alpha^*)}_{\text{Amenity Deduction}}$$

Appendix: Full Wage Premium Regressions

[Back](#)

	log(Wage)		
	(1)	(2)	(3)
Remote Work Intensity (α)		0.569*** (0.031)	0.351*** (0.035)
Squared Remote Work Intensity (α)		-0.535*** (0.071)	-0.221** (0.074)
Post-Pandemic Indicator $\times$ Remote Work Intensity (α)		0.036 (0.035)	0.218*** (0.039)
Post-Pandemic Indicator $\times$ Squared Remote Work Intensity (α)		0.081 (0.072)	-0.210** (0.077)
Teleworkability Index (ψ)	0.597*** (0.024)		0.522*** (0.026)
Squared Teleworkability Index (ψ^2)	-0.351*** (0.032)		-0.314*** (0.033)
Post-Pandemic Indicator $\times$ Teleworkability Index (ψ)	-0.127*** (0.023)		-0.178*** (0.025)
Post-Pandemic Indicator $\times$ Squared Teleworkability Index (ψ^2)	0.264*** (0.033)		0.272*** (0.034)
Year Fixed Effects	Yes	Yes	Yes
State Fixed Effects	Yes	Yes	Yes
Occupation Fixed Effects	Yes	Yes	Yes
Observations	359,708	359,708	359,708
R^2	0.417	0.419	0.421
R^2_{Within}	0.138	0.141	0.143
Individual Controls:	Yes	Yes	Yes